



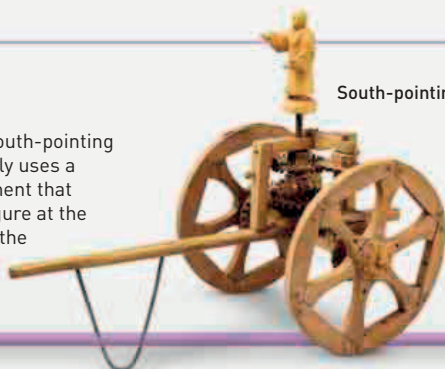
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THE NUMBER OF **GEAR WHEELS** FOUND IN THE REMAINS OF THE **ANTIKYTHERA MECHANISM** RECOVERED FROM A MEDITERRANEAN **SHIPWRECK**

c.230 BCE

Early gears

The Chinese south-pointing chariot probably uses a gear arrangement that ensures the figure at the front points to the south as the wheels turn.



South-pointing chariot

c.125 BCE

Zhang Heng's armillary sphere
Chinese scholar Zhang Heng builds an armillary sphere driven by gears and water. His model, showing the motions of the Sun, Moon, and stars, would go on to influence not only Chinese gear technology, but also later clockmakers (see 700–799).



Clock at Salisbury Cathedral

13th century

Mechanical clocks in Europe

The first mechanical clocks using gears to drive the rotation of pointers and to control the striking of chimes are invented.

They are powered by the controlled drop of a weight attached to a drive chain.



Chinese watermill

c.200 BCE

Watermills

Greek watermills that use gears to harness hydropower begin to spread throughout the Graeco-Roman world. The Chinese develop their own water wheel technology about 200 years later, with gearing mechanisms to drive various motions.



Windmill gear

c.7th century

Persian windmills

The first functional windmills, developed in Persia, have horizontal sails that drive the rotation of a vertical shaft.

1206

Book of Ingenious Devices

Arab polymath al-Jazari writes a treatise describing the construction of 100 remarkable machines—including the crankshaft—many of which rely on gears.

Al-Jazari's treatise

